

C PROGRAMMING MCQ TEST 3 – LOOPS

Question 1:

Which loop construct is best suited when the number of iterations is known?

- (a) for loop
- (b) while loop
- (c) do-while loop
- (d) switch case

Correct Answer: (a) for loop

Question 2:

What will be the output of the following C code?

```
#include <stdio.h>
void main()
{
    int x = 0;
    while (x++ < 3)
        printf("%d", x);
}
```

- (a) 012
- (b) 123
- (c) 111
- (d) 1234

Correct Answer: (b) 123

Question 3:

What will be the output of the following C code?

```
#include <stdio.h>
void main()
{
    int k = 0;
    for (k)
        printf("Hello");
}
```

- (a) Compile time error
- (b) Hello
- (c) Nothing
- (d) Varies

Correct Answer: (a) Compile time error

Question 4:

What will be the output of the following C code?

```
#include <stdio.h>
void main()
{
    double k = 0;
    for (k = 0.0; k < 3.0; k++)
        printf("Hello");
}
```

- (a) Run time error
- (b) Hello is printed thrice
- (c) Hello is printed twice
- (d) Hello is printed infinitely

Correct Answer: (b) Hello is printed thrice

Question 5:

What will be the output of the following C code?

```
#include <stdio.h>
void main()
{
    int k = 0;
    for (; k < 3; k++)
        printf("firstbit");
}
```

- (a) Compile time error
- (b) firstbitfirstbitfirstbit
- (c) Nothing
- (d) firstbitfirstbitfirstbitfirstbit

Correct Answer: (b) firstbitfirstbitfirstbit

Question 6:

Find the output of the given program.

```
#include <stdio.h>
void main()
{
    int i;
    for (;;)
    {
        printf("Hello fbs");
        break;
    }
}
```

- (a) Hello fbs
- (b) Hello fbs is printed infinite times
- (c) Compiled Successfully, No Output
- (d) Compile-time error

Correct Answer: (a) Hello fbs

Question 7:

Find the output of the given program.

```
#include <stdio.h>
int main()
{
    int i;
    for (printf("Hello "); printf("firstbit"); printf("solutions"))
    {
        break;
    }
    return 0;
}
```

- (a) Hello
- (b) Hello firstbit
- (c) Hello firstbit solutions
- (d) Compile-time error

Correct Answer: (b) Hello firstbit

Question 8:

What will be the output of the following C code?

```
#include <stdio.h>
void main()
{
    while (printf("Hello") - 5)
        printf(" World");
}
```

- (a) Nothing will print
- (b) Hello
- (c) Run time error
- (d) Infinite loop

Correct Answer: (b) Hello

Question 9:

What will be the output of this code?

```
void main()
{
```

```

int a = 1;
while (a <= 3 && printf("X "))
{
    printf("Y ");
    a++;
}

```

- (a) X Y X Y X Y X
- (b) X X Y
- (c) Infinite loop
- (d) X Y X Y X Y

Correct Answer: (d) X Y X Y X Y

Question 10:

What will be the output of this code?

```

void main()
{
    int i;
    for (i = 1; printf("N"); i++)
    {
        if (i > 5)
        {
            break;
        }
    }
}

```

- (a) N
- (b) NNNN
- (c) NNNNNN
- (d) Compile-time error

Correct Answer: (c) NNNNNN

Question 11:

What is the output of C Program?

```

void main()
{
    int k, j;

    for (k = 1, j = 10; k <= 5; k++)
    {
        printf("%d ", (k + j));
    }
}

```

- (a) Compiler error
- (b) 10 10 10 10 10
- (c) 11 12 13 14 15
- (d) None of the above

Correct Answer: (c) 11 12 13 14 15

Question 12:

What will be the output of the following C code?

```
#include <stdio.h>
void main()
{
    int i = 0;
    while (i++ < 2)
        while (i++ < 4)
            printf("%d ", i);
}
```

- (a) 2 3
- (b) 1 2 3 4
- (c) 2 3 4
- (d) 3 4

Correct Answer: (c) 2 3 4

Question 13:

What will be the output of the following C code?

```
#include <stdio.h>
void main()
{
    int x = 10;
    while (x-- >= 8)
        printf("%d ", x);
}
```

- (a) 10 9 8
- (b) 9 8 7
- (c) 10 9 8 7
- (d) 9 8 7 6

Correct Answer: (b) 9 8 7

Question 14:

What will be the output of the following C code?

```
#include <stdio.h>
void main()
{
```

```

    int i = 0;
    for (;;)
        printf("In for loop\n");

    printf("After loop\n");
}

```

- (a) Compile time error
- (b) In for loop After loop
- (c) In for loop infinite times
- (d) Undefined behaviour

Correct Answer: (c) In for loop infinite times

Question 15:

What will be the output of the following C code?

```

#include <stdio.h>
void main()
{
    int a = 1;
    while (printf("FBS%d", a) && a++ < 3)
        printf(" Java\n");
}

```

- (a) FBS1 Java FBS2 Java FBS3 Java
- (b) FBS1 Java FBS2 Java FBS3
- (c) Compile time error
- (d) FBS1 Java FBS3

Correct Answer: (b) FBS1 Java FBS2 Java FBS3

Question 16:

What will be the output of the following C code?

```

#include <stdio.h>
void main()
{
    while (printf("Firstbit Solutions\n"))
        break;
}

```

- (a) Firstbit Solutions
- (b) Firstbit Solutions (infinite times)
- (c) Compilation Error
- (d) No output

Correct Answer: (a) Firstbit Solutions

Question 17:

What is the purpose of the continue statement in a loop?

- (a) To exit the loop immediately
- (b) To skip the current iteration and move to the next iteration
- (c) To terminate the program
- (d) To execute a specific block of code

Correct Answer: (b) To skip the current iteration and move to the next iteration

Question 18:

Which statement is used to exit a loop prematurely?

- (a) return statement
- (b) continue statement
- (c) break statement
- (d) exit statement

Correct Answer: (c) break statement

Question 19:

What is the output of the following code snippet?

```
void main()
{
    int x = 0;
    while (++x && x < 3)
        printf("x=%d\n", x);
}
```

- (a) x=1 x=2
- (b) x=0 x=1 x=2
- (c) x=1 x=2 x=3
- (d) x=0

Correct Answer: (a) x=1 x=2

Question 20:

What is the output of the following program?

```
void main()
{
    int i = 0;
    for (i = 0; i < 10; i++)
    {
        continue;
    }
    printf("%d", i);
}
```

- (a) 10
- (b) 0
- (c) Compilation Error
- (d) 9

Correct Answer: (a) 10

Question 21:

What is the output of the following program?

```
void main()
{
    int i;
    for (i = 0; i < 10; i++)
    {
        if (i % 2 == 0)
        {
            continue;
        }
        printf("%d", i);
    }
}
```

- (a) Program will print all even numbers between 0 to 10
- (b) The program will print all odd numbers between 0 to 10
- (c) Program gives a compilation error
- (d) None of the above

Correct Answer: (b) The program will print all odd numbers between 0 to 10

Question 22:

Find the output of the given program.

```
#include <stdio.h>
void main()
{
    int i = 3;
    while (i-- && printf("A "))
        printf("B ");
}
```

- (a) A B A B A B
- (b) A A B B
- (c) A B A
- (d) A A B B A A B B

Correct Answer: (a) A B A B A B

Question 23:

What is the output of the given below program?


```
#include <stdio.h>
void main()
{
    int i = 0;
    while (i < 5)
    {
        i++;
        if (i == 3)
            continue;
        printf("%d ", i);
    }
}
```

- (a) 1 2 3 4 5
- (b) 1 2 4 5
- (c) 1 2 3 5
- (d) 0 1 2 4 5

Correct Answer: (b) 1 2 4 5

Question 24:

What is the output of the given below program?

```
#include <stdio.h>
int main()
{
    int i = 10;
    while (i-->0)
    {
        if (i == 5)
            break;

        if (i % 2 == 0)
            continue;

        printf("%d ", i);
    }
}
```

- (a) 9 7 5
- (b) 9 7
- (c) 9 8 7 6
- (d) 9 7 6

Correct Answer: (b) 9 7

Question 25:

What is the output?

```
#include <stdio.h>
int main()
{
```

```

    int n;
    for (n = 9; n != 0; n--)
        printf("%d", n--);

    return 0;
}

```

- (a) 9 7 5 3 1
- (b) 9 8 7 6 5 4 3 2 1
- (c) Infinite Loop
- (d) 9 7 5 3

Correct Answer: (a) 97531

Question 26:

What will be the output of the following program?

```

#include <stdio.h>
int main()
{
    int i;
    for (i = 1; i != 10; i += 2)
        printf("Firstbit ");

    return 0;
}

```

- (a) Firstbit Firstbit Firstbit Firstbit
- (b) Firstbit infinite times
- (c) Firstbit Firstbit Firstbit Firstbit
- (d) Firstbit Firstbit Firstbit Firstbit Firstbit Firstbit

Correct Answer: (b) Firstbit infinite times

Question 27:

What's going to happen when we compile and run the following C program?

```

#include <stdio.h>
int main()
{
    int j = 0;

    for (; j < 10;)
    {
        if (j < 10)
            printf("Firstbit", j++);
        else
            continue;

        printf(" Solutions\n");
    }
}

```

```
        return 0;
    }
```

- (a) Compile Error due to use of continue in for loop
- (b) No compile error but it will run into infinite loop printing Solutions
- (c) Compile error
- (d) It'll print Firstbit Solutions 10 times.

Correct Answer: (d) It'll print Firstbit Solutions 10 times.

Question 28:

What will be the output of the following C code?

```
#include <stdio.h>
void main()
{
    int a = 0;

    while (a < 5)
    {
        if (++a == 3)
            break;

        printf("%d ", a);
    }
}
```

- (a) 1 2
- (b) 1 2 3
- (c) 0 1 2
- (d) Compilation error

Correct Answer: (a) 1 2

Question 29:

What is the output of the following C code?

```
#include <stdio.h>
void main()
{
    for (printf("CSK\n"); 0; printf("KKR\n"))
        printf("MI\n");
}
```

- (a) CSK MI
- (b) CSK KKR MI
- (c) CSK
- (d) Compilation error

Correct Answer: (c) CSK

Question 30:

What is the output of the following C code?

```
#include <stdio.h>
void main()
{
    int a = 4;
    int sum = 0;
    int i;

    for (i = 1; i < a; i++)
    {
        sum = sum + a;
    }

    printf("The value of sum and a are %d and %d respectively", a, sum);
}
```

- (a) The value of sum and a are 10 and 4 respectively
- (b) The value of sum and a are 12 and 4 respectively
- (c) The value of sum and a are 4 and 12 respectively
- (d) The value of sum and a are 4 and 10 respectively

Correct Answer: (c) The value of sum and a are 4 and 12 respectively